The Vertical Datum, Gravity & Geoid

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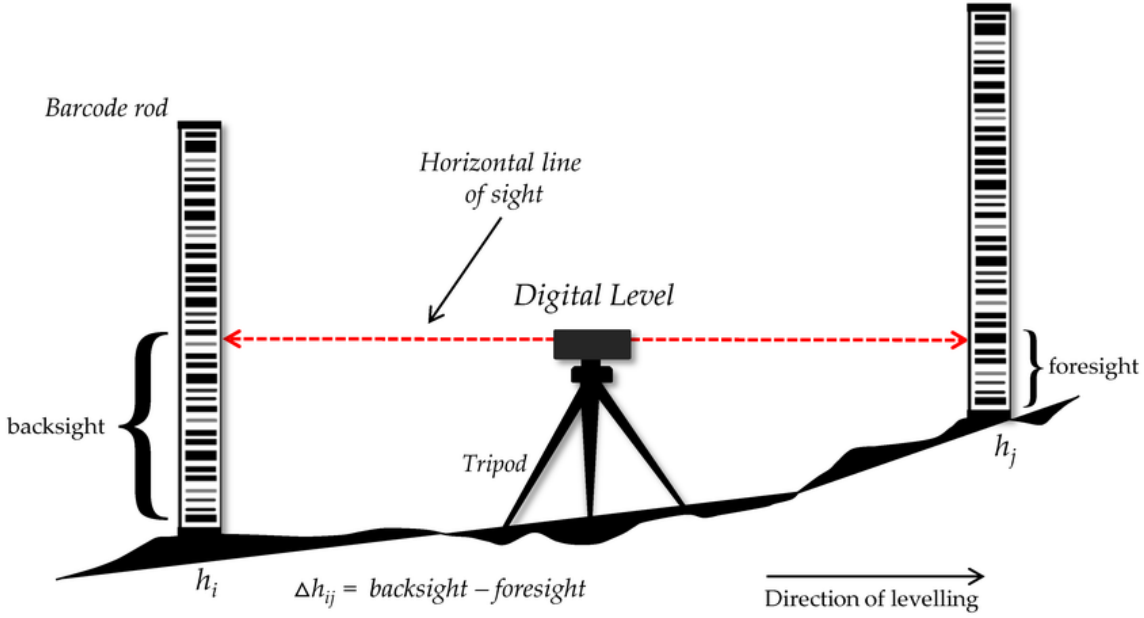
Brief Overview

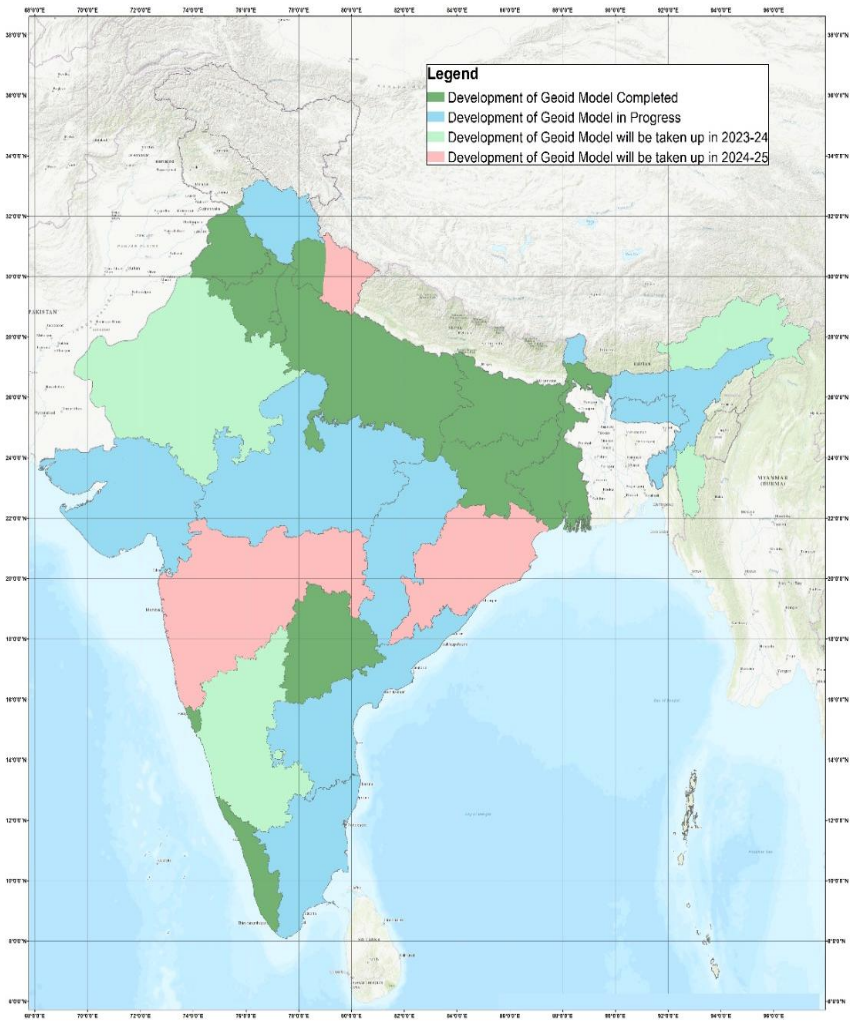
- **Vertical Datum**: A vertical datum is a reference surface used to measure elevations or depths. In India, the Mean Sea Level (MSL) serves as the vertical datum, and it provides a consistent frame of reference for height measurements.



- **Gravity Survey**: A gravity survey measures the gravitational field at different points on Earth's surface. The variations in gravity help determine mass distribution within the Earth, contributing to various geophysical studies and geoid modeling.

- **Geoid Models**: The geoid is a hypothetical surface representing mean sea level across the globe. It is crucial for determining precise elevations using satellite-based systems like GNSS.

Importance in Geodesy and Mapping



- **Geodesy**: Vertical Datums, gravity surveys, and geoid models are essential for understanding the Earth's shape, gravity field, and how these influence the surface. They ensure accurate geodetic measurements and coordinate systems.

- **Mapping Applications**: They are vital for producing precise maps, especially in large infrastructure projects, navigation, and flood management, where accurate height information is crucial for planning and analysis

List of Height Types

Heights above or below a reference surface

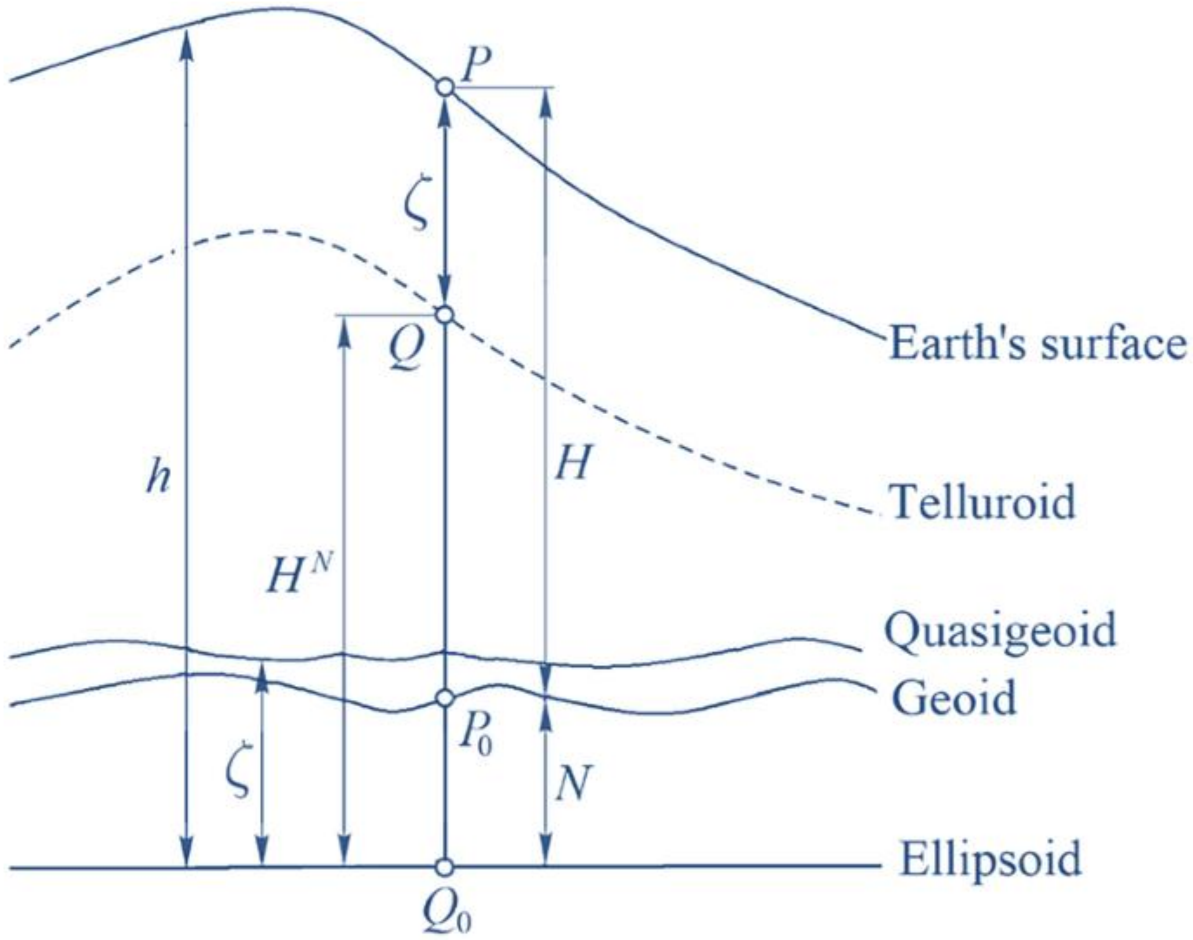
- Ellipsoidal Height (h)
- Orthometric Height (H)
- Normal Height (H^N)
- Dynamic Height (H^D)
- Geoidal Height / Geoid Undulation (N)
- Quasigeoidal Height / Height Anomaly (ζ)
- Physical Height
- Geopotential Height
- Helmert Orthometric Height
- Normal-Orthometric Height
- Spherical Height
- Astronomical Height
- Tidal Heights
  - Mean Tide Height
  - Zero Tide Height
  - Tide-Free Height
- Sea Surface Height (SSH)
- Mean Sea Level Height (MSL)
- Chart Datum Height
- Depth / Sounding Height (negative height)

Why Do We Need Different Height Systems?

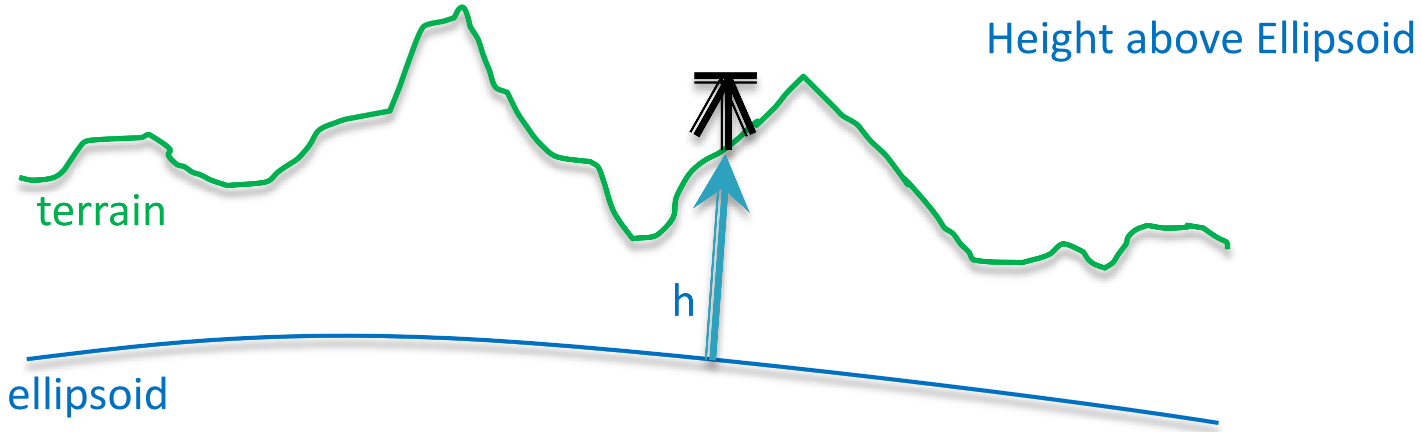
- Height is not purely geometric
- Earth's gravity field is non-uniform
- Engineering, hydrology and GNSS require different interpretations of height
- Vertical datum depends on equipotential surfaces

Reference Surfaces for Heights

- Ellipsoid (mathematical surface)
- Geoid (equipotential surface ≈ MSL)
- Quasigeoid
- Equipotential surfaces of gravity field
- Telluroid
- Earth Surface



1. Ellipsoidal Height (h)



Positive is UP, negative is down

The ellipsoidal height is the distance measured along the normal (perpendicular) to the ellipsoid from the point to the reference ellipsoid.

2. Geopotential Numbers

Concept

- Fundamental quantity in vertical positioning
- Represents **potential difference** between point and reference surface
- Basis for all physical height systems

Definition

$$C = W_0 - W_P$$

Where

- W_0 = Gravity potential of reference surface (geoid)
- W_P = Gravity potential at point

Use (India)

- Precise levelling adjustment
- Vertical datum definition
- GNSS–geoid–levelling integration

3. Orthometric Height (H)

Concept

- True **physical height above the geoid** (MSL)
- Measured along the **plumb line**
- Depends on Earth's **actual gravity field**

Formula

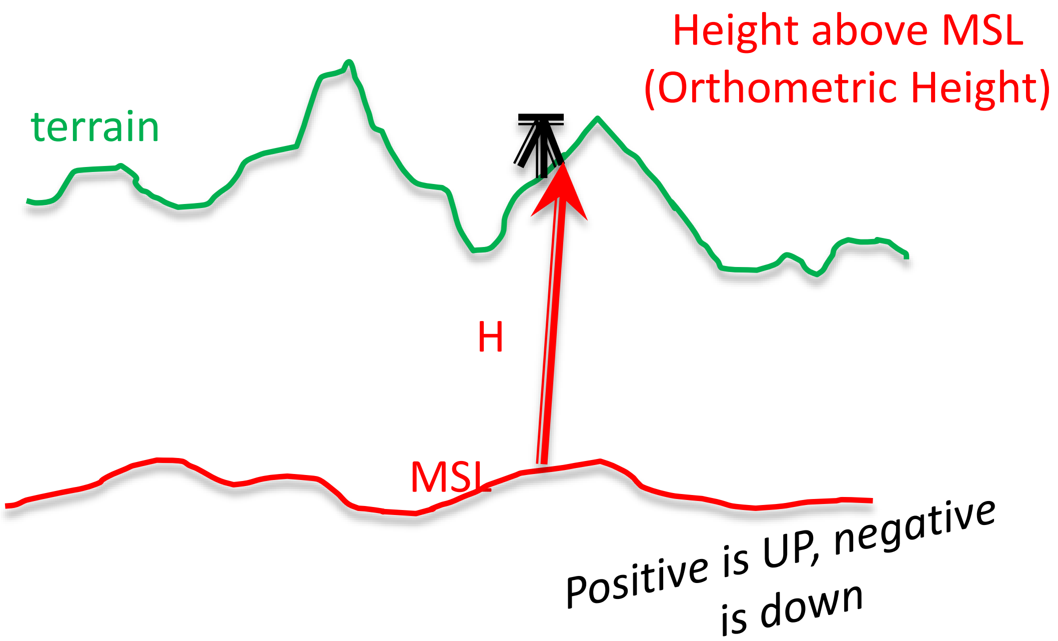
$$H = \frac{C}{\bar{g}} = \frac{W_0 - W_P}{\bar{g}}$$

Where

- C = Geopotential number
- $\bar{g}$ = Mean actual gravity along plumb line

Use (India)

- Survey of India levelling
- Engineering, mapping and infrastructure
- National vertical datum (classical)



Normal Gravity (γ)

Concept

- Gravity of an **ideal, rotating reference ellipsoid**
- Free from local mass anomalies
- Used as a **theoretical reference** in height systems

Formula (Somigliana's Formula)

$$\gamma(\varphi) = \frac{\gamma_e\,(1 + k \sin^2 \varphi)}{\sqrt{1 - e^2 \sin^2 \varphi}}$$

Where

- φ = Geodetic latitude
- γ_e = Normal gravity at equator
- e = First eccentricity of ellipsoid
- k = Gravity constant of the ellipsoid

Use (India)

- Computation of **normal heights**
- Reduction of levelling observations
- GNSS–geoid–vertical datum integration

4. Normal Height (H_n)

Concept

- Height above **quasigeoid**
- Uses **normal gravity field**
- Avoids Earth density problem

Formula

$$H^N = \frac{C}{\bar{\gamma}} = \frac{W_0 - W_P}{\bar{\gamma}}$$

Where

- C = Geopotential number
- $\bar{\gamma}$ = Mean normal gravity

Use (India)

- Large-area levelling when gravity data unavailable
- Vertical datum modernisation
- GNSS–geoid integration

5. Dynamic Height (H_o)

Concept

- Physical height based on **geopotential difference**
- Preserves **equipotential surfaces**
- Same height $\Rightarrow$ same gravity potential

Formula

$$H^D = \frac{C}{\gamma_0} = \frac{W_0 - W_P}{\gamma_0}$$

Where

- C = Geopotential number
- γ_0 = Constant normal gravity

Use (India)

- Hydrology and river gradient studies
- Historical levelling (pre-gravimeter era)
- Research, not routine surveying

Indian Context Summary

Height Type	Formula	Gravity Used
Orthometric	$H = \frac{C}{\bar{g}}$	Mean actual gravity
Normal	$H^N = \frac{C}{\bar{\gamma}}$	**Mean normal gravity**
Dynamic	$H^D = \frac{C}{\gamma_0}$	Constant normal gravity

Period	Height Practice in India
Pre-gravimeter era	Normal / normal-orthometric heights
Early hydrological studies	Dynamic heights
Modern levelling (SoI)	Orthometric / normal-orthometric
GNSS era	$h + N \rightarrow H$

6. Tidal Datum (H_t)

- Tidal datum is typically based on mean sea level or other tidal references.
- The **tidal datum** is typically based on the mean sea level (MSL) or another tidal reference, like the **Mean High Water** (MHW) or **Mean Low Water** (MLW).
- For example, **Mean Sea Level** is commonly used as a reference:

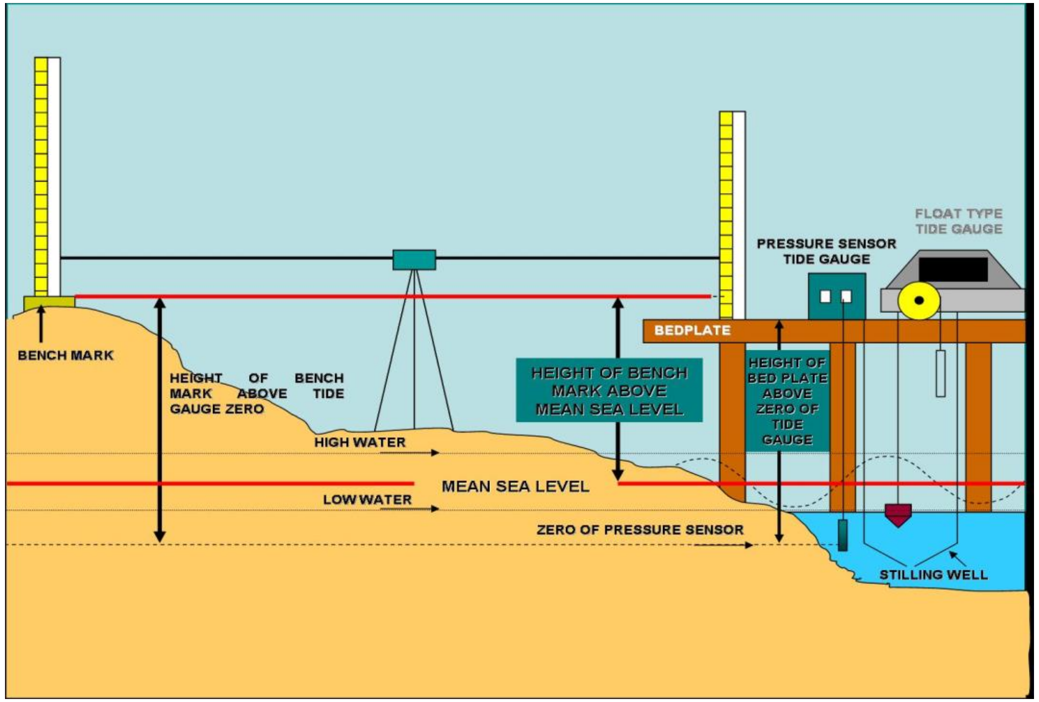
$$H_t = \text{Mean Sea Level} - \text{Reference Level}$$

- The exact formula depends on the tidal reference used and the region.

MSL HEIGHTS

Heights above or below a reference surface

- **Mean Sea Level Heights**
- **Ellipsoidal Heights**
- **Orthometric Heights**



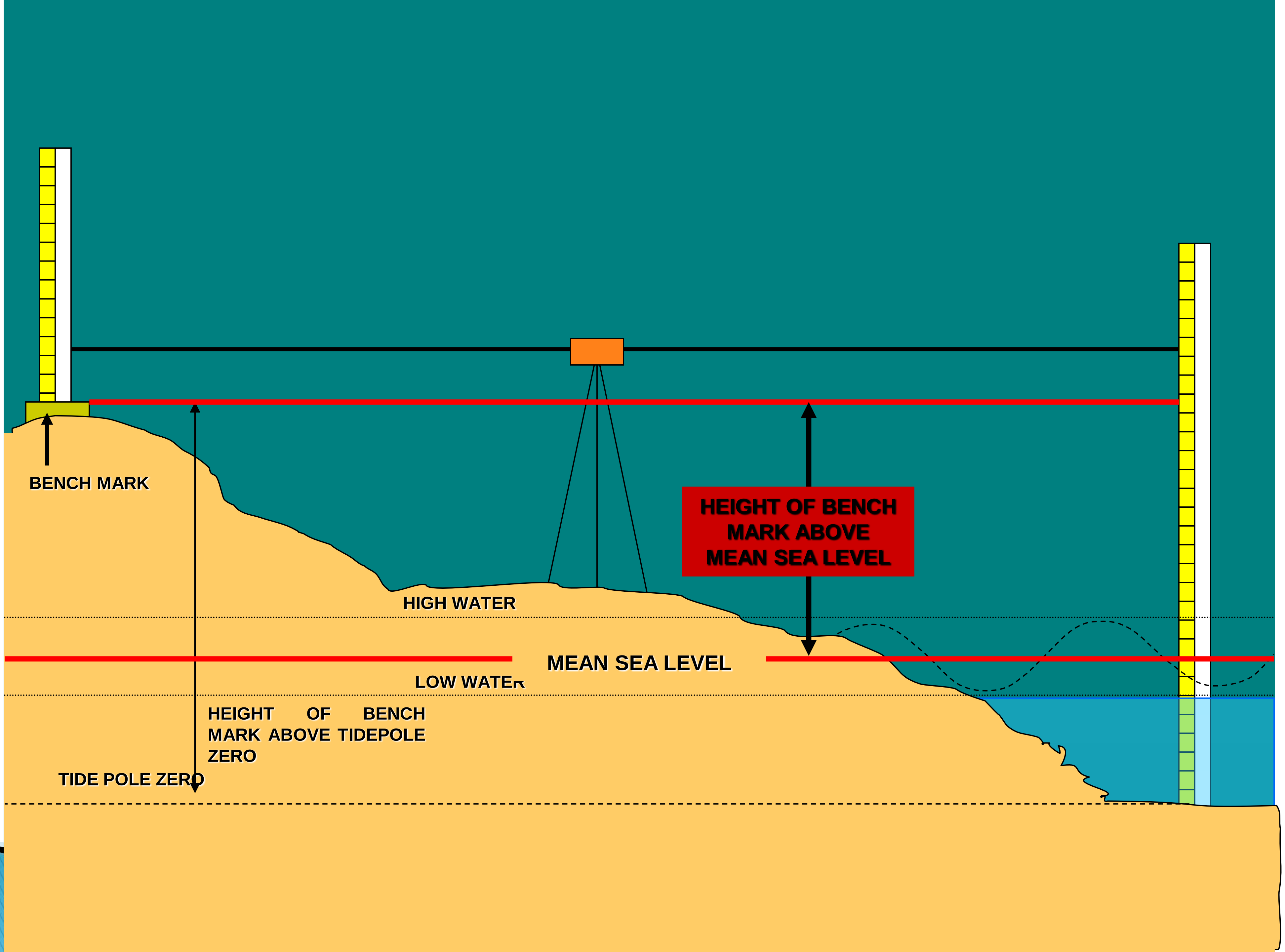
The distance measured along a perpendicular, between a point and a reference surface.

Height: those that employ the earth's gravity field as their datum - **"Orthometric height"**

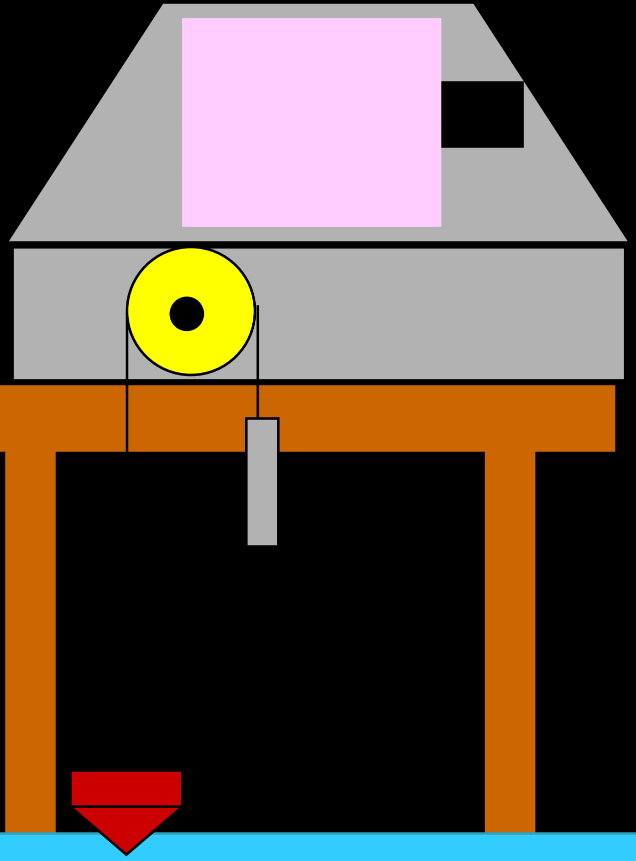
It is now known that mean sea level can differ from geoid by up to a metre or more, but the exact difference is difficult to determine.

For fixing the Vertical datum for country, a set of 18.61 years as the period is suitable for measurement of mean sea level at tide gauge.

The choice of 18.61 year was chosen because it is the smallest integer number of years larger than the first major cycle of the moon's orbit around the earth.


BENCH MARK
HEIGHT OF BENCH MARK ABOVE MEAN SEA LEVEL
HIGH WATER
MEAN SEA LEVEL
LOW WATER
HEIGHT OF BENCH MARK ABOVE TIDEPOLE ZERO
TIDE POLE ZERO

OBSERVATIONS USING ANALOGUE TIDE GAUGES



FIRST LEVEL NET of IVD

1875: Gen. Walker initiated vertical datum operations.

1909: Operations completed.

Tidal Observatories: Data from nine observatories (4 east coast, 4 west coast, 1 in Karachi) used.

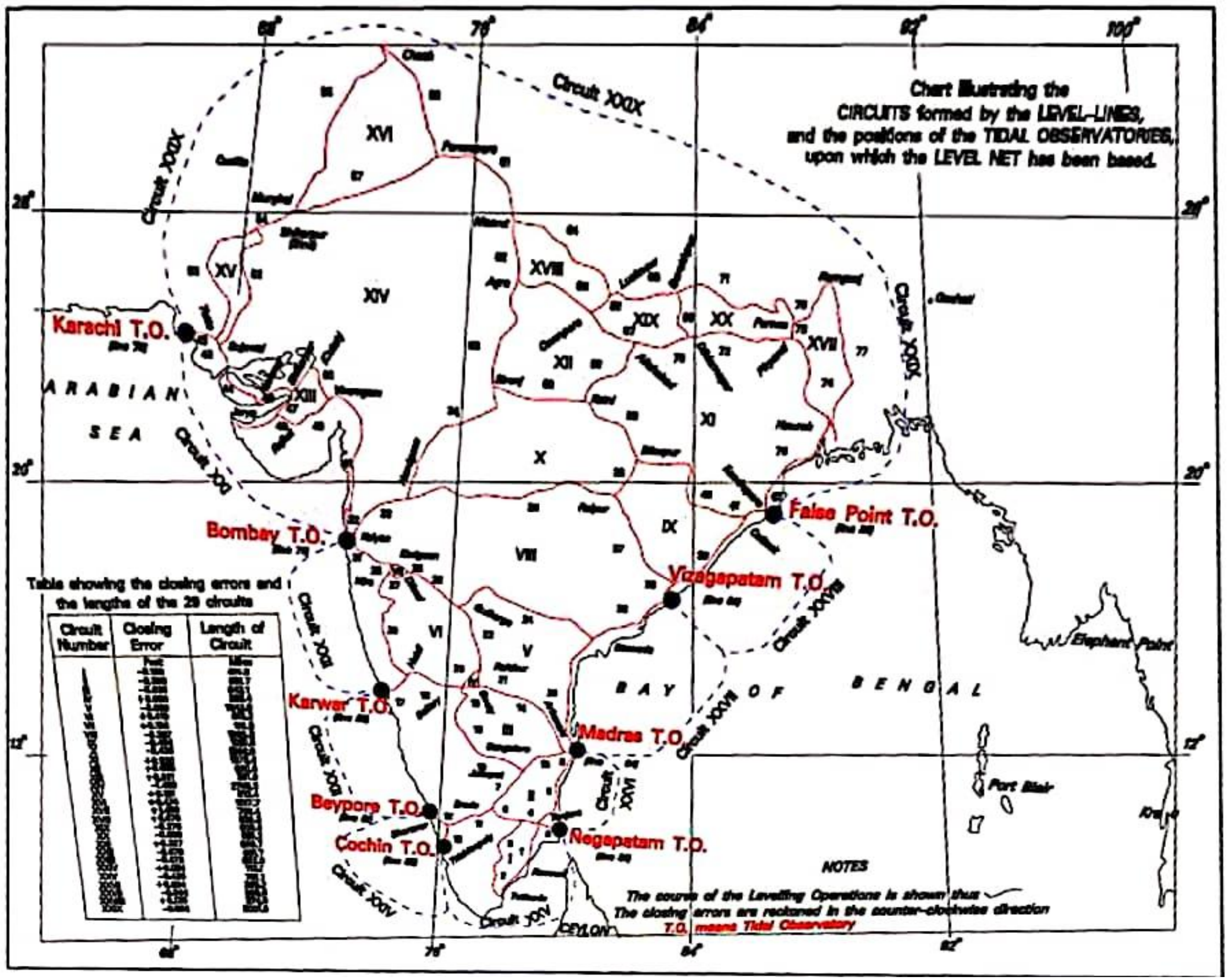
Data Duration: Tidal data ranged from 4 to 40 years.
Assumption: Sea levels at all observatories were considered equal to fix the vertical datum.

No Actual Gravity: The level network did not use actual observed surface gravity for Orthometric height calculations.
Normal Gravity: Instead, approximate gravity (normal gravity) from a mathematical model was used.
Dynamic Correction: Derived using normal gravity values based on benchmark latitudes.
Standard Latitude: Dynamic corrections were calculated assuming a standard latitude of 24° for the Indian subcontinent, with the datum point at latitude 24°.

WHY MODERNIZATION OF IVD REQUIRED ?

- M.S.L. of all the ports was considered at the same level.
- Change in MSL value.
- No fore and back levelling concept old Instruments and staves with no calibration.
- Inconsistency in level network Adjustment.
- No observed gravity for orthometric correction.
- 95% of leveling bench marks(BMs) destroyed.
- No relationship with GNSS derived positions.
- Not suitable for construction of geoid model of cms level accuracy.

FIRST LEVEL NETWORK 1909



GCP Library

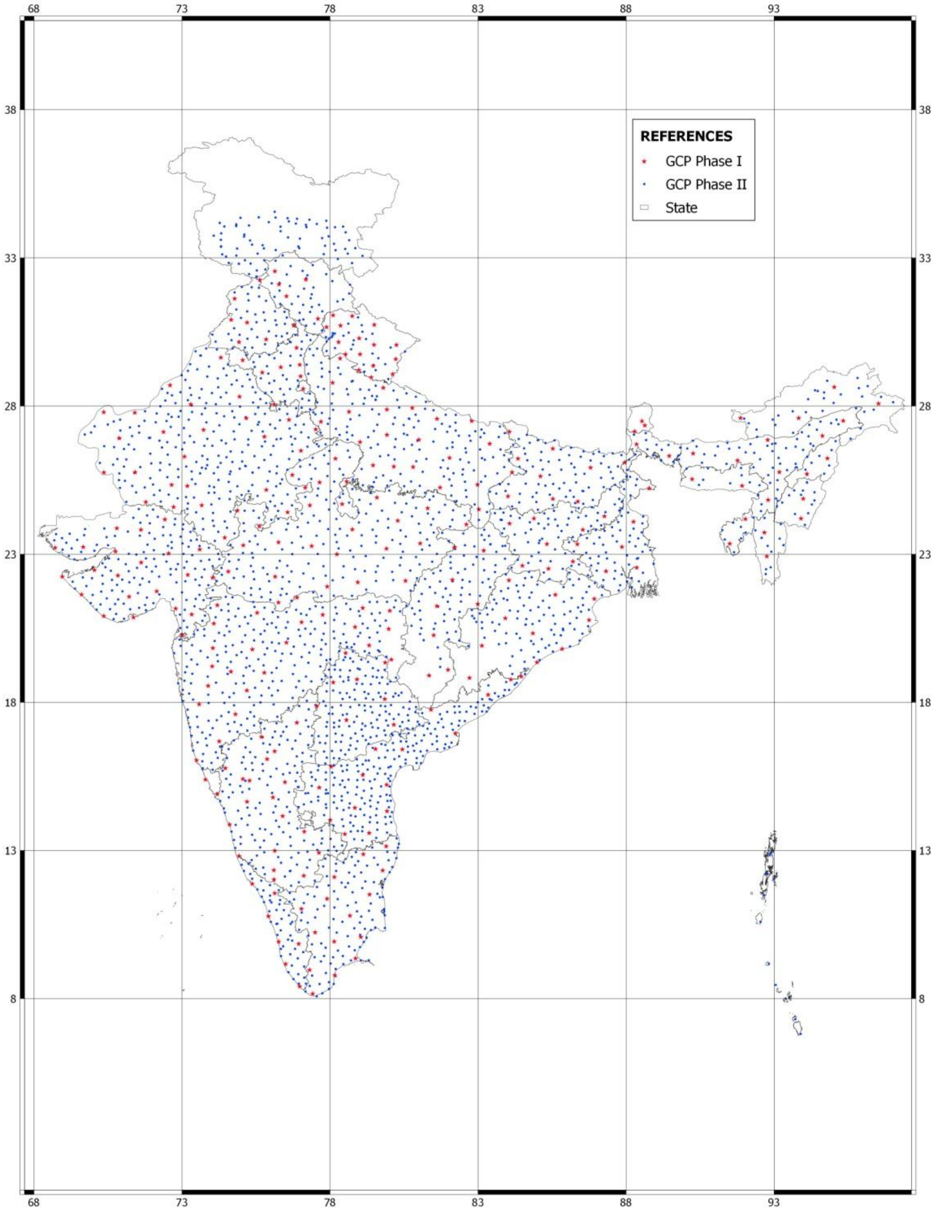
•**Satellite Technology**: Mapping transitioned to a **Geo-centric Coordinate System** (e.g., GRS80, WGS84).

•**Survey of India** Shifted from the **Everest Spheroid** to **Geo-centric Coordinate System (ITRF)**.

Ground Control Points (GCPs):

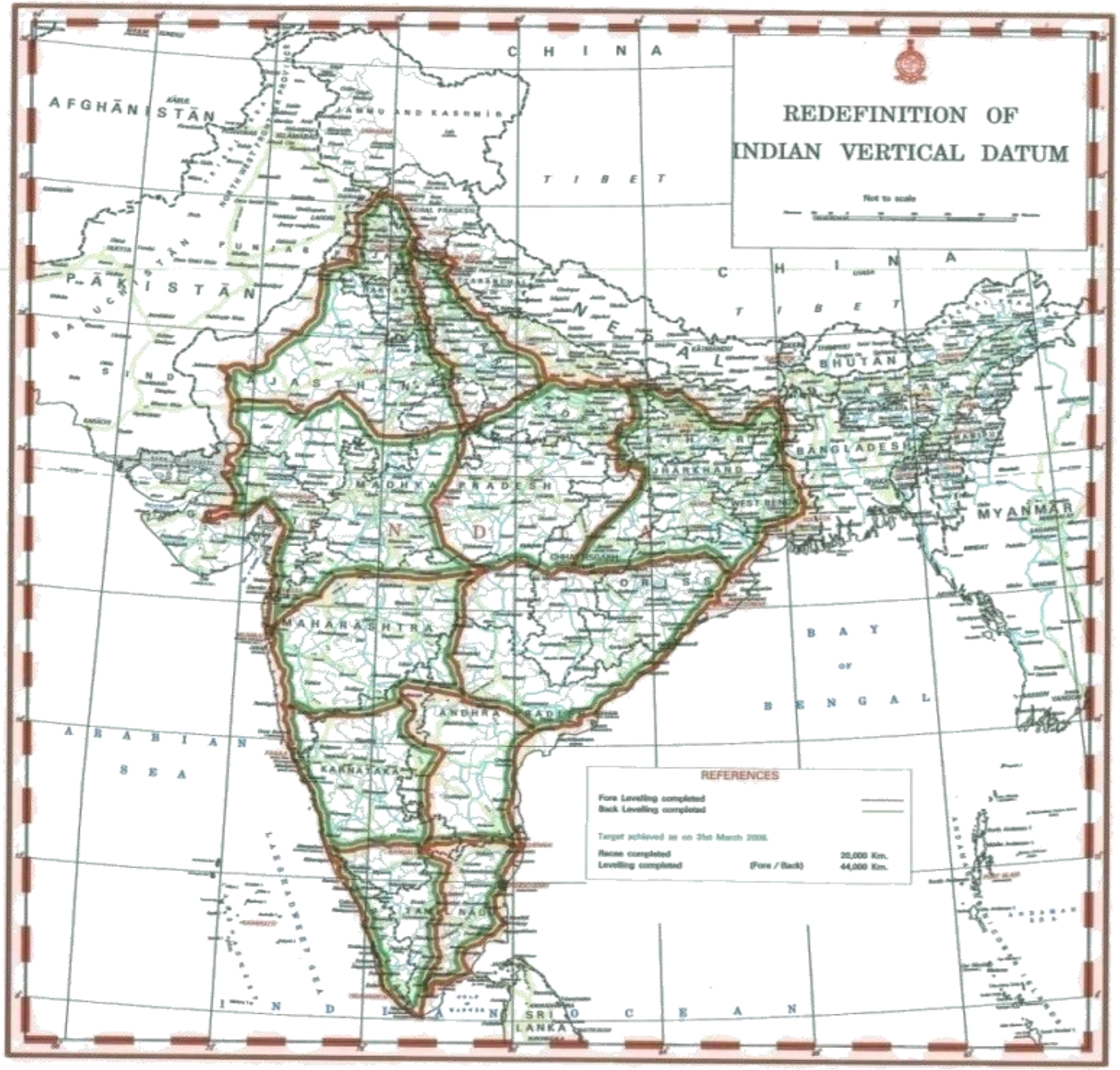
•**First Phase**: 260 GCPs spaced 250-300 km apart.

•**Second Phase**: 2260 GCPs spaced 25-30 km apart.



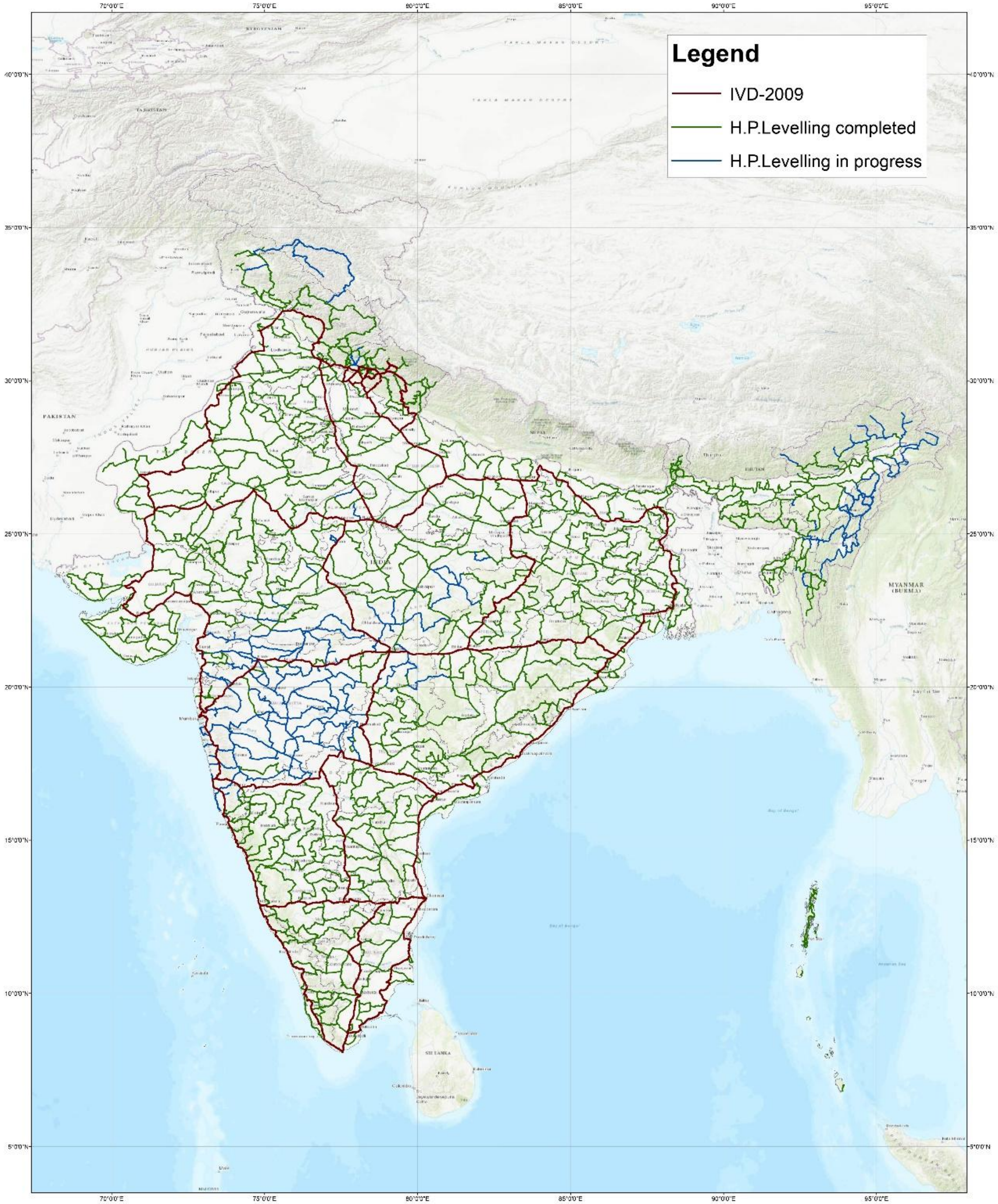
High Precision Levelling:

- **19,417 km** of levelling using N-3, Trimble Dini-12, and Leica DNA03 instruments.
- 29 levelling lines forming 19 junction points and 11 circuits.
- **Vertical Datum Reference**: Indian Mean Sea Level (IMSL) from Apollo Bunder Tidal Observatory.
- **Surface Gravity:** Measured every 3rd to 5th benchmark, assuming linear variation.
- **Geo-potential Number:** Calculated and summed for each levelling line.
- **Least Square Adjustment:** Network adjusted with geo-potential values for 19 junction points.
- **Mean Gravity:** Calculated using Poincare and Prey method.
- **Helmert Orthometric Height:** Computed post-adjustment.



Spirit Levelling Data

Approx. 2,00,000 Kms of levelling in fore & back direction has been completed



GEODETIC DATUMS

HORIZONTAL DATUM

- The ability of GNSS to derive fairly precise and accurate horizontal positions is well known and fully established.
- CORS

VERTICAL DATUM

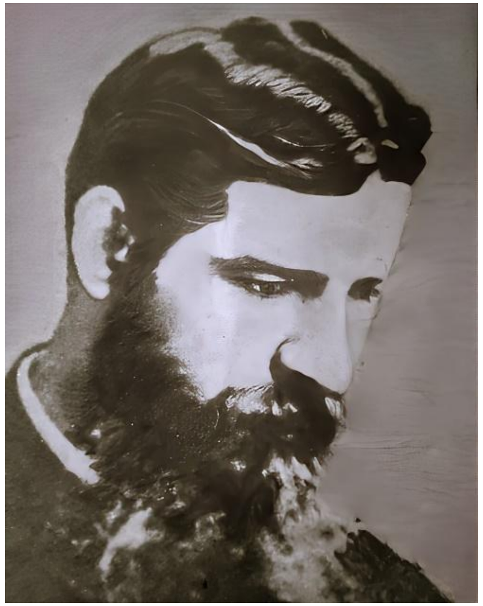
- Heights above a level surface

  1. Levelling – Tidal Datum

  2. Geoid – Gravity Datum

GRAVITY

- The Knowledge of Gravity Field is very important in the study of our globe. If we know the gravity anomalies we can determine the figure of the earth and shape of the Geoid.
- Survey of India is carrying out Gravity observations since 1865. Initially the pendulum Gravimeters were used.
- After world war II spring based relative gravimeters are in use.
- Recently One Absolute Gravimeter is also been procured.



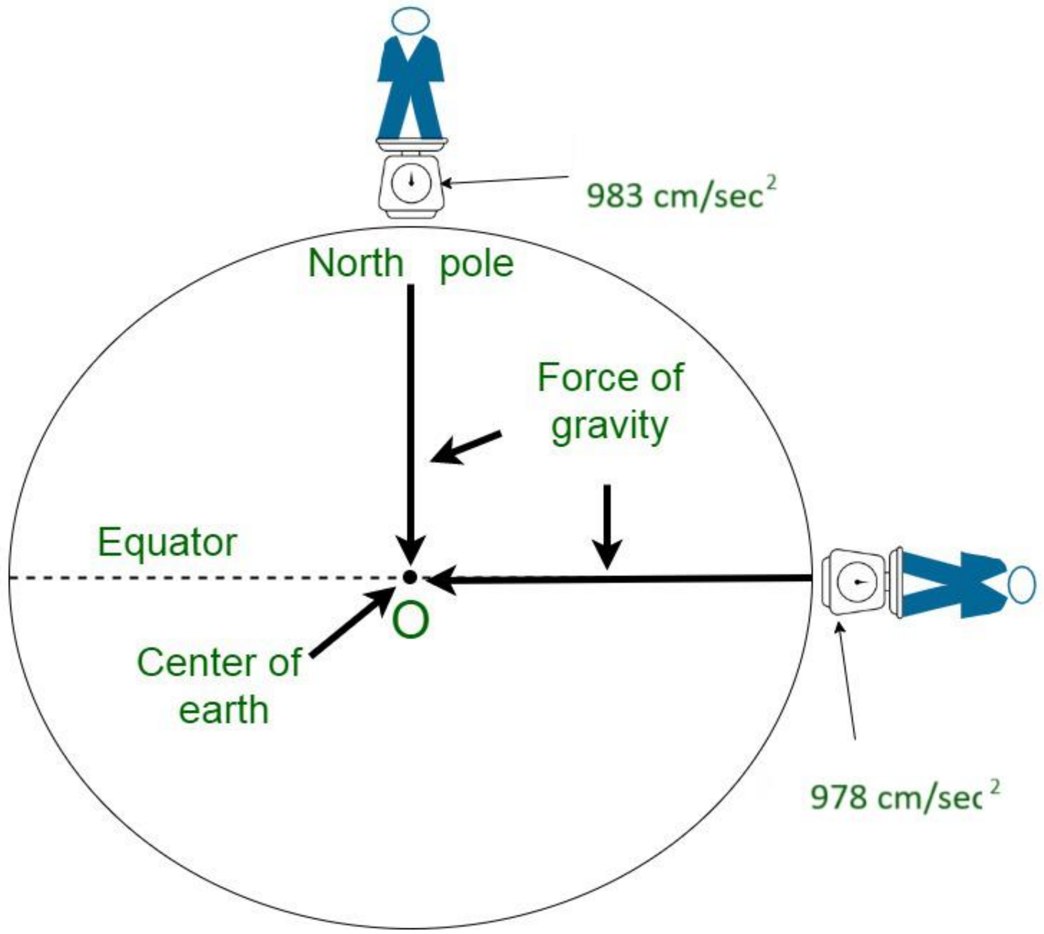
Variation in the value of ‘g’

On surface of the earth ‘g’ is
minimum at equator and maximum at poles.

- Shape of the earth
- Axial rotation of the earth

- ‘g’ decreases on going above the surface of the earth.

- ‘g’ decreases on going below the surface of the earth.

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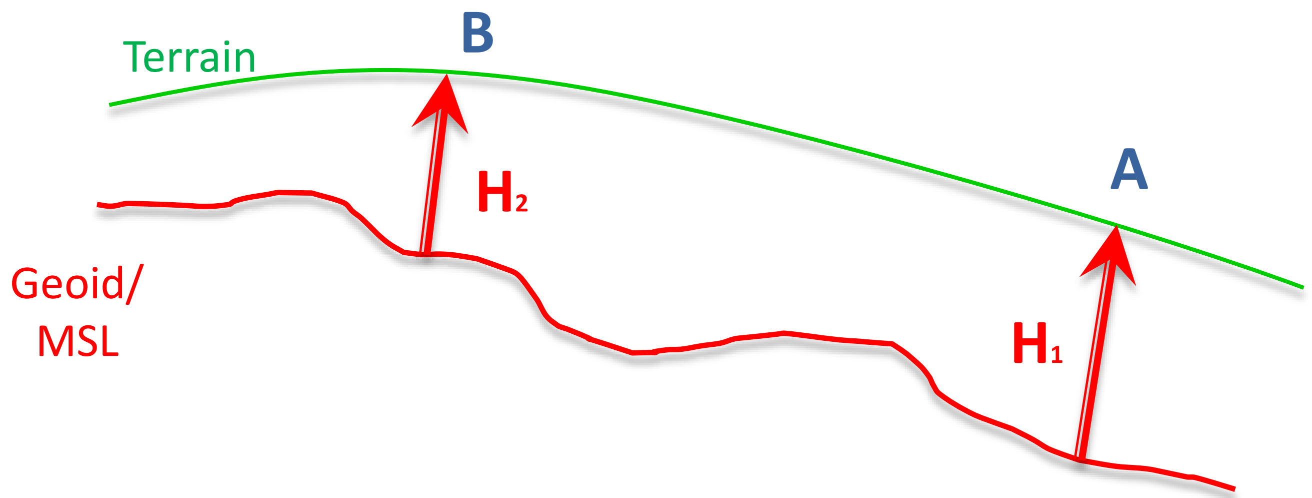
Why water flows from down to up in some places?

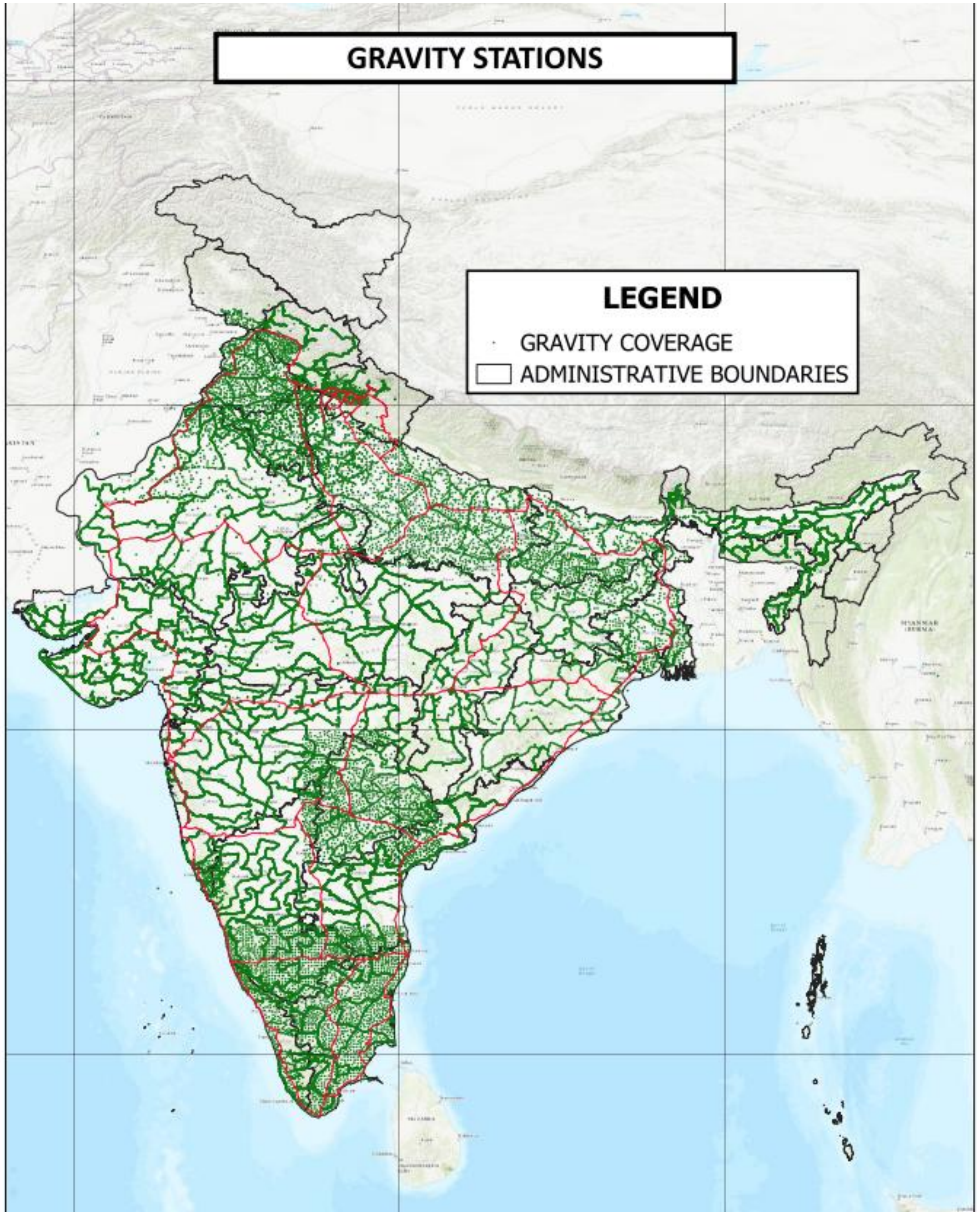
Orthometric Height above geoid is more at A



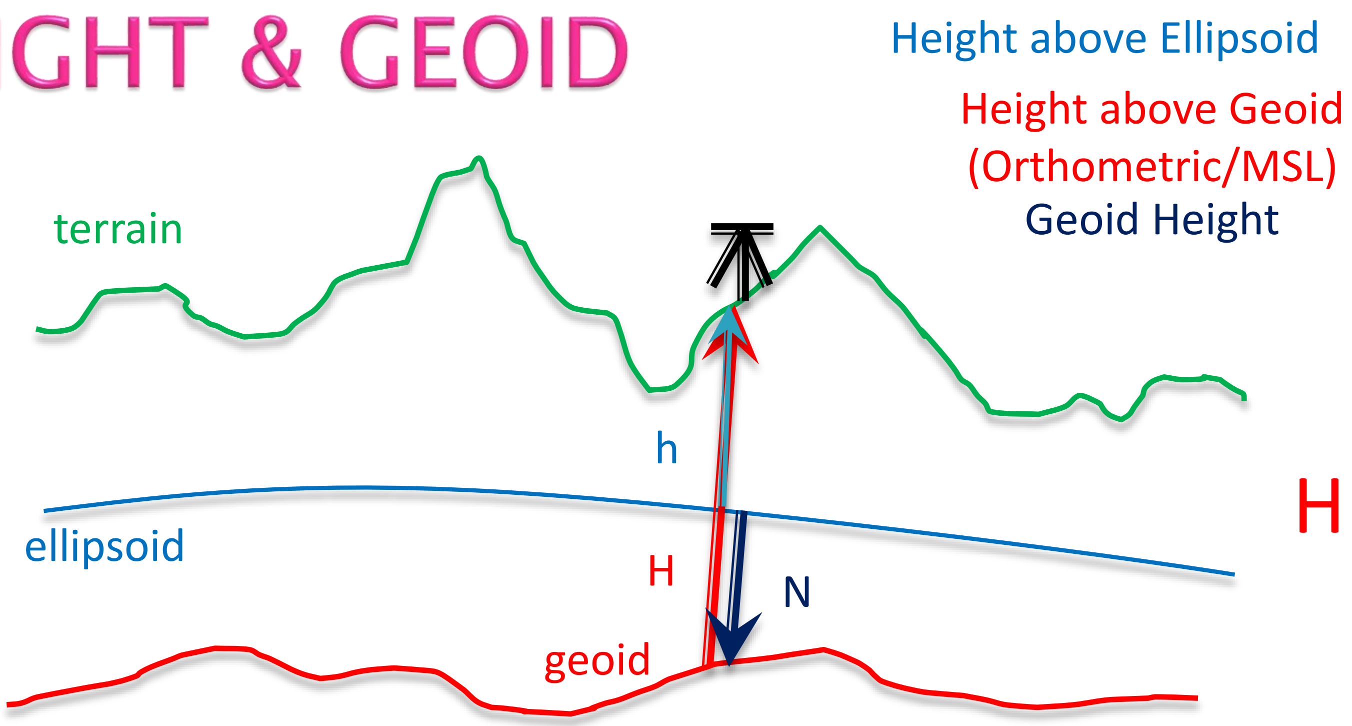
The values of Gravity are used in the computations of Orthometric Heights.

Gravity Coverage in India

50000 Stations



HEIGHT & GEOID



$$H = h - N$$

The derived ellipsoidal height (h) is not suitable for practical purposes as it doesn't follow the principle of water flow.

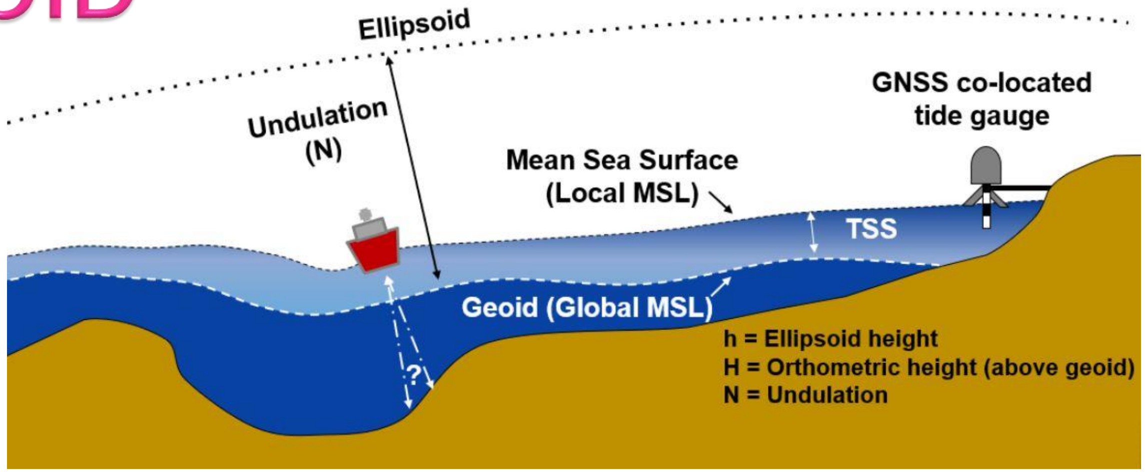
It means that if we can make a model of values of 'Geoidal Undulations (N)',

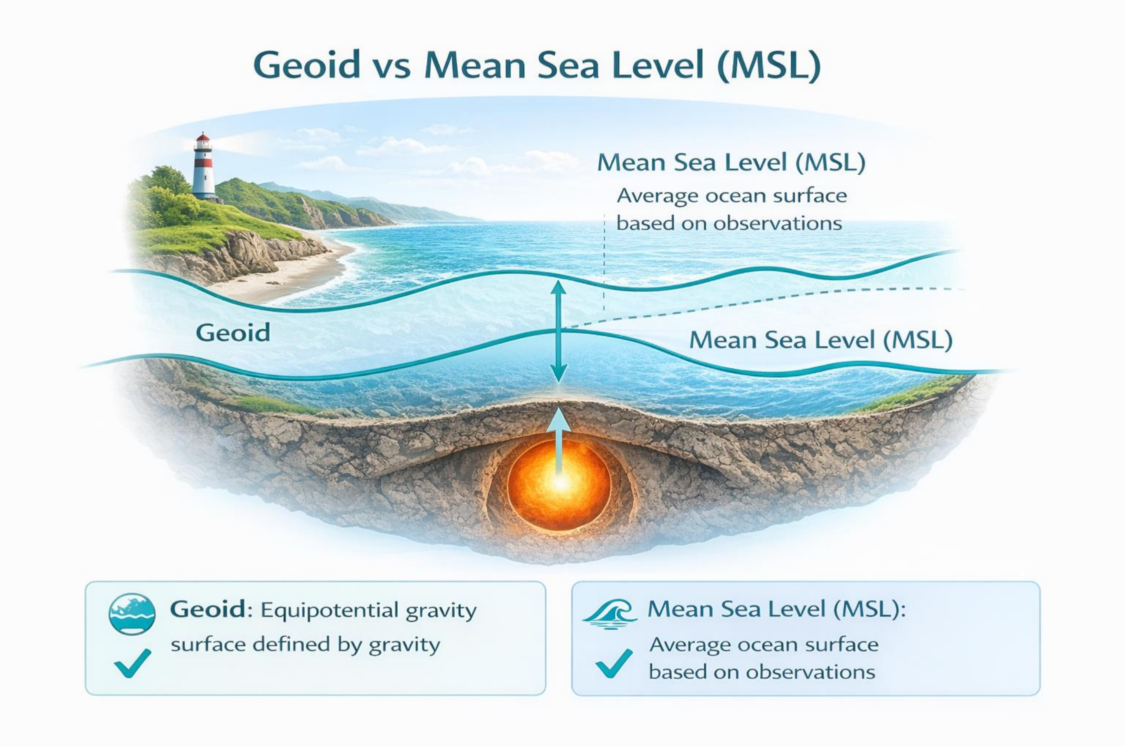
We can calculate values of 'H' using Ellipsoidal heights (h) obtained from GNSS Receivers.

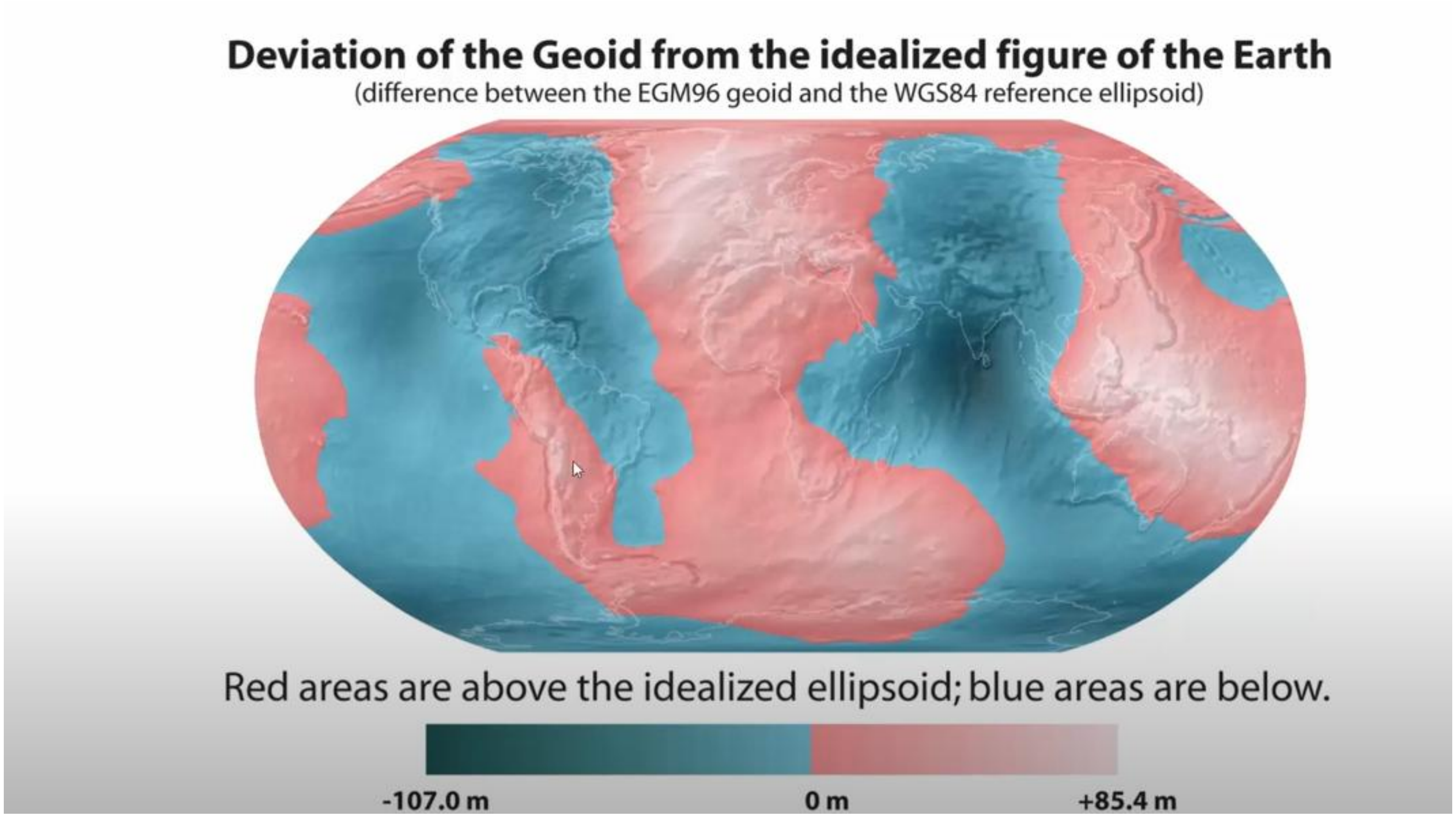
This Model is called Geoid Model.

MSL & GEOID

Geoid	MSL
Equipotential gravity surface	Mean ocean surface
Defined by gravity ((W = const))	Defined by observations
Extends under land & sea	Only over oceans
Stable, time-invariant	Varies with time
Reference for orthometric height	Tide-gauge reference

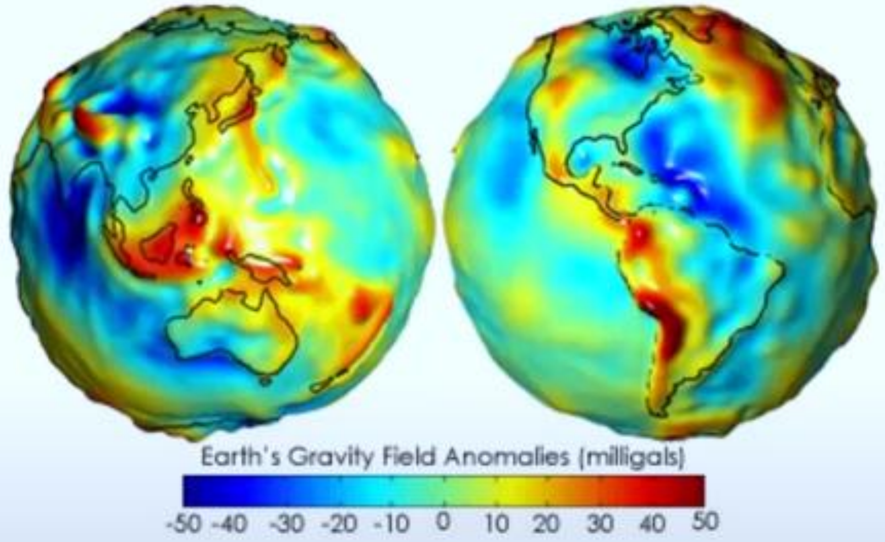




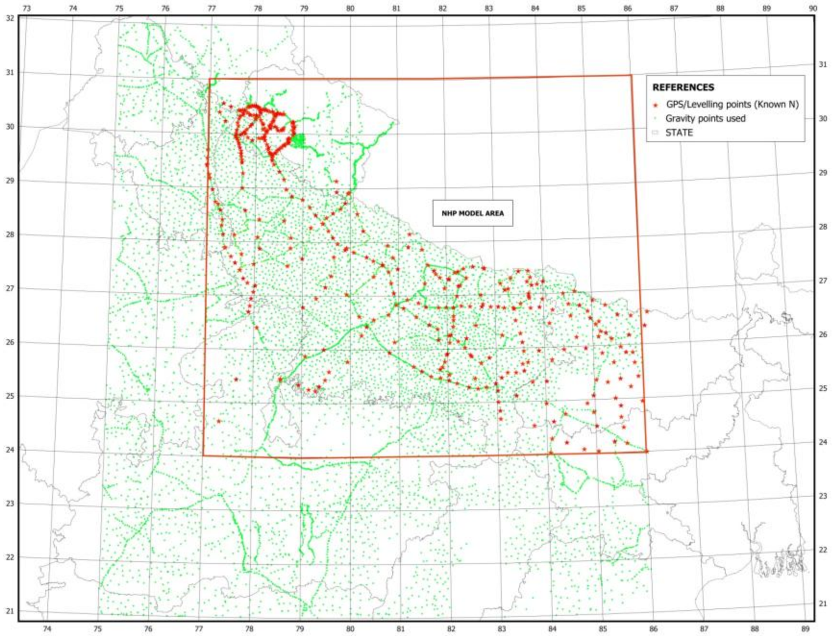

Deviation of the Geoid from the idealized figure of the Earth
(difference between the EGM96 geoid and the WGS84 reference ellipsoid)
Red areas are above the idealized ellipsoid; blue areas are below.
-107.0 m
0 m
+85.4 m

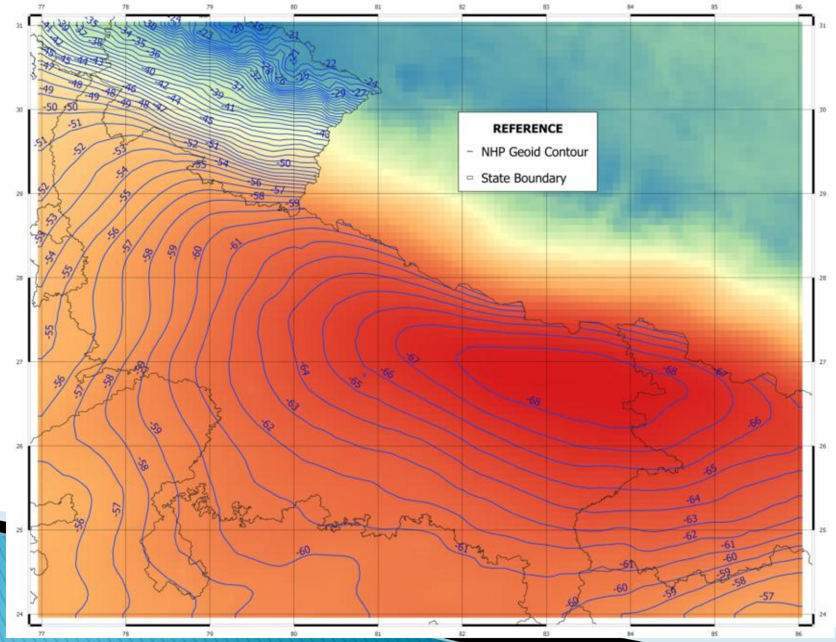
Global geoid models

- EGM08
- EGM96
- EGM2020
  - Global coverage
  - Lower accuracy



Regional Geoid (UP & Bihar)





- Terrestrial gravity data i.e. well distributed free air anomaly in an around the area of interest.

- Well distributed data of GNSS observations on Bench Marks connected to the IVD 2009 in the area.

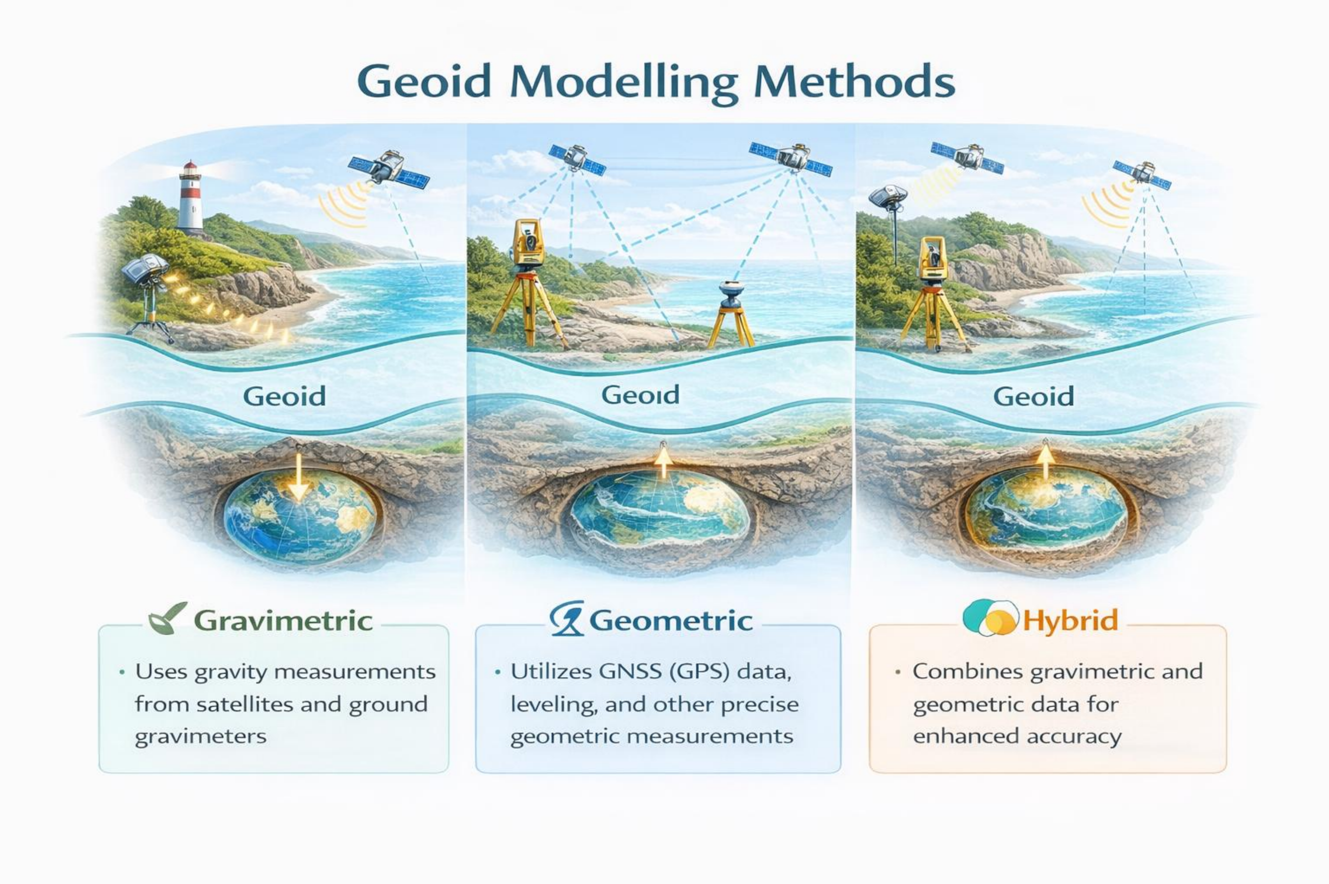
N = h – H

Comparision between observed N & NHP Hybrid Geoid N

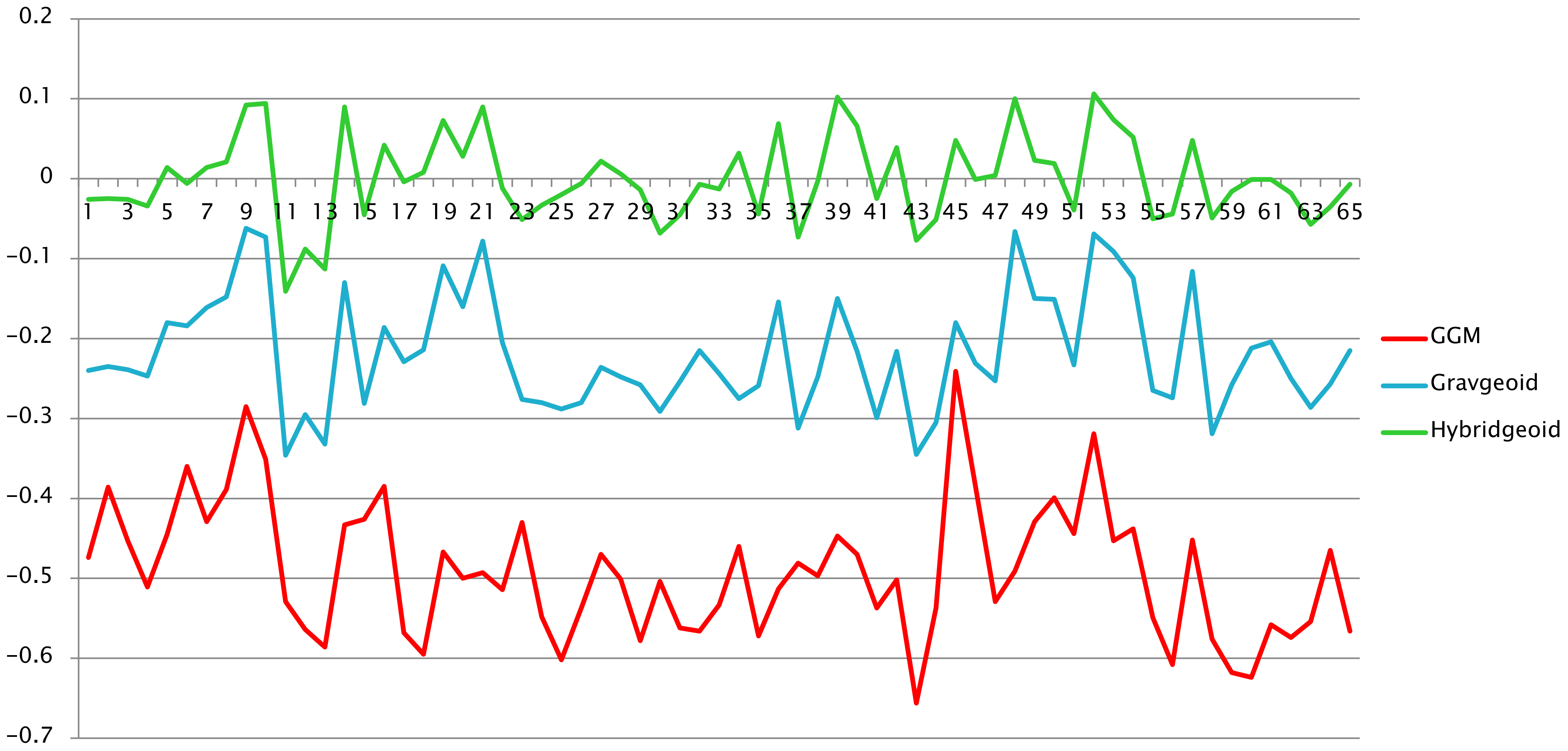
RMSE	Maximum difference	Minimum difference	Average	STD
0.078	0.212	-0.198	-0.0111	0.0768

GEOID MODELLING METHODS

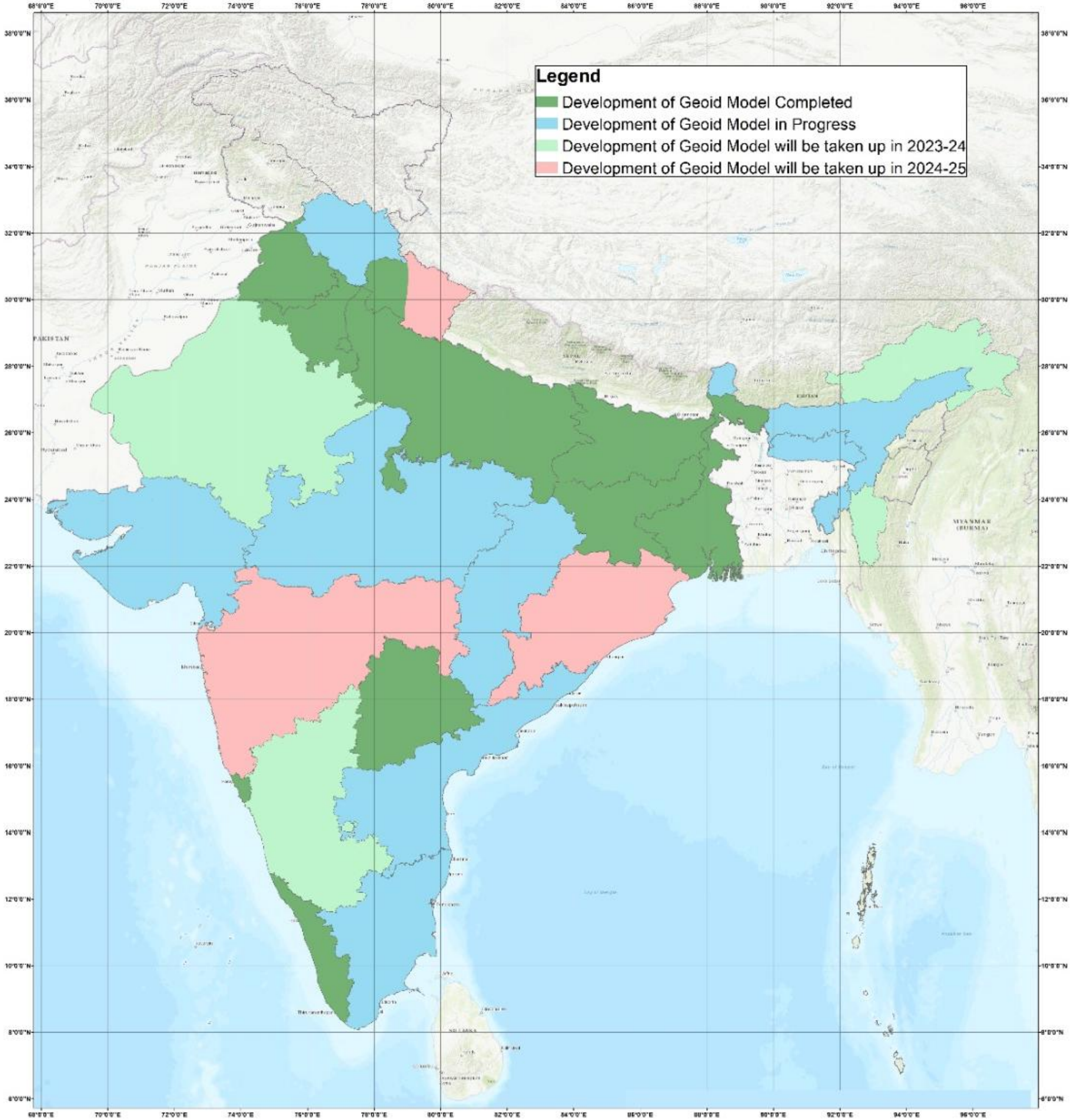
- **Gravimetric** approach mainly applied for global and regional models.
- **Geometric** approach for determination of local/regional geoid with an aim to replace levelling measurements with GNSS surveys.
- A combination of above two techniques commonly known as **hybrid** approach of geoid determination.



Difference between Global & Regional Geoid



Status of Geoid Model in India



Use of Geoid Model

1. Precision and Accuracy
2. Real-time Monitoring
3. Surveying and Mapping
4. Flood Modeling and Prediction
5. Groundwater Management
6. Infrastructure Planning and Monitoring
7. Optimizing Water Use
8. Natural Resource Management

1. To obtain Orthometric heights from GNSS receivers directly.
2. To convert an ellipsoidal DEM in to Orthometric DEM.

Different Formats of Geoid Models

- Understanding different formats of geoid models is essential for working with geodetic data in various applications.
- Choose the appropriate format based on the requirements of your project and the compatibility with your software and tools.

We can convert Geoid Models

- Trimble: Trimble Grid Factory
- Leica: Geoid Model Reader
- AGISOFT: GEOTIFF
- ERDAS: Binary Format (Have inbuilt option)
- ARCGIS: ASCII GRID, GEOTIFF
- QGIS: ASCII GRID, GEOTIFF
- Hydromagic: Geoid Converter Utility
- Global Mapper: Have option to convert in different formats
- PIX4D: only uses Global Model or a Shift

Future Plans

- Recently Survey of India has developed high resolution Geoid models for the projects like NHP, NMCG, SWAMITVA etc.

- At Present Geoid Model of UP, Bihar Jharkhand, Haryana, Punjab, Goa, West Bengal, Telangana, Kerala and Part of Uttarakhand State are available.

- More data collection for development of more precise IndGeoid is already under way. The Survey of India is expected to release Geoid Model for rest of India shortly.

- It will definitely facilitate GNSS user of any community to derive faster & cheaper MSL heights.

Conclusion

- IVD which is now all set to come up
- will provide an authoritative height system to avoid confusion.
- Users can obtain heights above the Geoid or a level equipotential) surface close to it.
- It will help the users to compute the local instantaneous sea level in terms of the national vertical datum.
- To provide the ability to correlate the local level (equipotential) surface and global ellipsoidal and geopotential surfaces.
- IVD will provide a reliable frame for consistent analysis and development of Gravimetric Geoid model for India.
- It will support unification of multiple vertical datums and authoritative transformations of heights to an acceptable and defined accuracy.

Thank you!

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